

Project Initialization and Planning Phase

Date	03 October 2026
Team ID	SWTID-2026-7859
Project Title	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal describes EduGenie, a generative-AI learning assistant that makes learning simpler, faster and more personal. It tackles the lack of instant, simplified and personalized help for students. Key features include Gemini-powered Q&A, concept explanation, quiz generation, summarization and learning-path recommendations.

Project Overview

Section	Description
Objective	Build an easy-to-use AI tutor that answers questions, explains concepts simply, generates quizzes, summarizes passages and recommends personalized learning paths using Google Gemini.
Scope	A web application (FastAPI backend + HTML/CSS/JS frontend) with five learning features, REST APIs, error handling, automated tests and local deployment. Out of scope for this version: user accounts, progress storage, voice and mobile app.

Problem Statement

Section	Description
Description	Students struggle with complex concepts, lack practice questions and revision aids, and don't know how to plan their learning; existing resources are scattered, lengthy and not personalized.
Impact	Solving this saves study time, improves understanding and exam confidence, and makes quality learning support accessible to anyone with a browser - even on low-end devices.

Proposed Solution

Section	Description
Approach	Use Google Gemini (via the google-genai SDK) with carefully engineered prompts for each learning task, a lightweight local model (LaMini-Flan-T5-783M) for explanations, FastAPI REST endpoints and a simple responsive web UI.

Section	Description
Key Features	- Q&A: concise answers to academic questions - Explain: simple explanations for school students - Quiz: MCQs (4 options) in validated JSON, interactive scoring - Summarize: overview + key bullet points - Learning Path: beginner to advanced plan with timelines and resources

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Any modern dual-core CPU (Intel / AMD / Apple M1); no GPU needed (AI runs on Google Cloud)
Memory	RAM specifications	4 GB (8 GB recommended if the local LaMini-Flan-T5 model is used)
Storage	Disk space for data, models, and logs	~500 MB (app + dependencies); +3 GB for the optional local model
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-genai, python-dotenv, pydantic, pytest, httpx; optional: transformers, torch, sentencepiece
Development Environment	IDE	Visual Studio Code (Python extension), Git, GitHub
Data		
Data	Source, size, format	No training dataset required - uses pre-trained models (Google Gemini via API; MBZUAI/LaMini-Flan-T5-783M from Hugging Face). Input: user text (JSON); output: text / JSON.